Assignment 4

A- 2D Array Tasks

1. Calculate the Sum of the Main Diagonal of a Matrix

Write a program that reads the elements of a square matrix and calculates the sum of the elements along its main diagonal (also known as the major diagonal).

Input:

Matrix elements:

1 2 3
4 5 6
7 8 9

Output:

Sum of main diagonal elements = 15

2. Add Two Matrices

Write a program to read the elements of two 3x3 matrices and calculate their sum. The program should output a new matrix, where each element is the sum of the corresponding elements from the two input matrices.

Input:

Matrix 1:

1 2 3
4 5 6
7 8 9

Matrix 2:

9 8 7
6 5 4
3 2 1

Output:

Sum of both matrices:

10 10 10
10 10 10
10 10 10

3. Calculate the Row Sum of a Matrix

Write a program that reads the elements of a matrix and calculates the sum of the elements in each row. The program should output the sum of elements for each row separately.

Input:

Matrix elements:

1 2 3
4 5 6
7 8 9

Output:

Sum of row 1 = 6
Sum of row 2 = 15
Sum of row 3 = 24

4. Check for a Symmetric Matrix

Write a program in C++ to check whether a given matrix is symmetric. A symmetric matrix is a square matrix that is equal to its transpose. The program should verify if the input matrix satisfies this condition.

Input 1:

Matrix elements:

1 2 3
2 4 5
3 5 8

Output 1:

The matrix is symmetric.

Input 2:

Matrix elements:

1 2 3
2 4 5
3 5 8

Output 2:

The matrix is asymmetric.

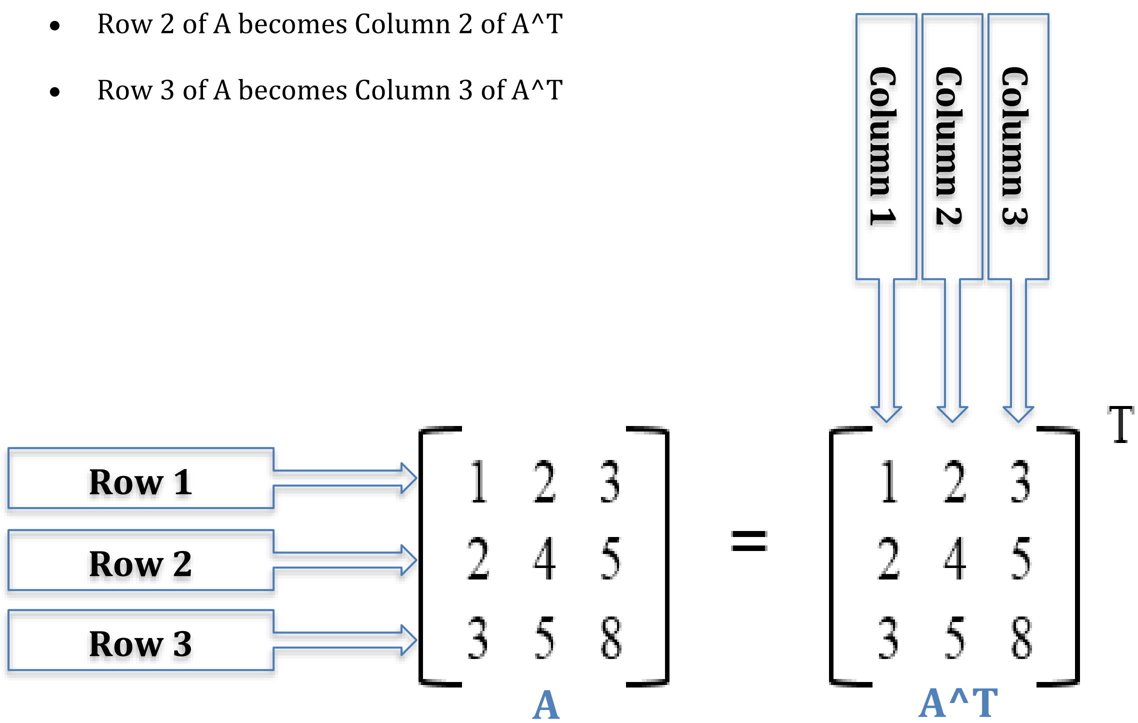
Definition of a Symmetric Matrix:
A symmetric matrix is a square matrix (number of rows equals number of columns) that is equal to its transpose. **For a symmetric matrix A:** A = A^T

How to get the Transpose of a Matrix:

The **transpose** of a matrix is obtained by flipping it over its diagonal. This means the rows of the original matrix become the columns in the transposed matrix, and the columns become the rows.

To transpose:

- Row 1 of A becomes Column 1 of A^T
- Row 2 of A becomes Column 2 of A^T
- Row 3 of A becomes Column 3 of A^T



Symmetric matrix

B- Function

5. Void Function to Print an Integer

Create a void function that takes an integer as a parameter and prints it.
Call this function in the `main` function.

Input: 5
Output: 5

6. Function to Calculate Average of Three Integers

Create a function that takes three integers as parameters, calculates their average, and returns it as a float.
Call this function in the `main` function to display the result.

(Note: Inputs can be separated by a space or by a new line i.e. hitting Enter)
Input: 5 3 2
Output: 3.33333

7. Function to Check if a Number is Prime

Create a function that takes an integer as a parameter and returns `true` if the number is prime, otherwise returns `false`.
Call this function in the `main` function to test various numbers.

Input: 7
Output: true

Input: 8
Output: false

8. Function to Check if a Number is Even

Create a function that takes an integer as a parameter and returns `true` if the number is even, otherwise returns `false`.
Call this function in the `main` function.

Input: 8
Output: true

Input: 7
Output: false

9. Find Maximum and Minimum Using Functions

Write a program to input an array of integers from the user **(You can assume the array size to be of any size more than 1).** Create a function to find the maximum and minimum of the given numbers.
Call this function to display the results in the `main` function.

Input: {20, 5, 30, 40, 50, 1}
Output:
Maximum: 50
Minimum: 1

10. Function to Find the Cube of a Number

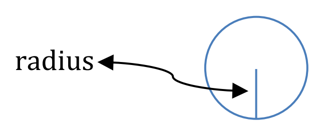
Create a function that takes a single integer as a parameter and calculates its cube.
Call this function in the `main` function.

(The cube is the result of multiplying a value by itself three times as follows -> 5 X 5 X 5 = 125)

Input: 5
Output Cube of 5 = 125

11. Circle Measurements Using Functions

Write a program to input the radius of a circle from the user. Create a function that takes the radius as a parameter, calculates and prints the diameter, circumference, and area of the circle. **(Assume π = 3.1416)**.
Call this function in the `main` function.



Input: Radius = 10
Output:
Diameter = 20 units
Circumference = 62.832 units
Area = 314.16 square units

12. Function to Print All Divisors of a Number

Create a void function that takes a number as input and prints all its divisors.
Call this function in the `main` function.

Input: 8
Output: 1 2 4 8

13. Function to Check Divisibility by 3 and 4

Create a function that takes a number as a parameter and returns `true` if the number is divisible by both 3 and 4; otherwise, it returns `false`.
Call this function in the `main` function to test various numbers.

Input: 12
Output: true

Input: 15
Output: false